

Ai OPS 2

Harshvardhan Agrawal

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Q1

Size of the naive docker image is 1.47GB, Multistage: 442MB, reduction of 70.64%

python:3.11-slim doesnt include the full debian build environment that are present in python3:11, resulting in 970MB savings

–no-cache-dir is what caused this difference of 56MB as the wheel pip leaves in the /root/.cache/pip in naive approach while multistage doesnt maintain it.

Slim runtime has no system libgomp but sklearn wheel has its own so it still works.

Q2

One dockerfile cannot make more than one image, so it cannot declare that our spam API needs a seerate redis container. Docker compose solves this by one declarative file, that defines the two services and their shared network, starting order, and configuration, and one docker compose up brings the whole machine up. Compose's scope is still a single docker host, with no scheduling across machines and no automatic reschedulin if the host gives up, whereas kubernetes orchestrates container across clusters, and can place pods by resource reqursts.